

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION-BASED QUESTIONS

Code of Conduct:

I K.Sujana sree(192525320) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

APPLICATION-BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

A) (a) Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

The Caesar Cipher is one of the oldest and simplest encryption techniques. In this method, every letter in the plaintext is shifted by a fixed number of positions in the alphabet. Here, the shift value is 4.

Step 1: Plaintext

“MEET AT NOON”

Step 2: Apply Shift +4 i.e; key=4;

Formula for Ceasar Cipher $E=(P+K) \bmod 26$

Plain Letter| Shift +4| Cipher Letter

M| M → N → O → P → Q| Q $M=12+4=16 - Q$

E| E → F → G → H → I| I $E=4+4=8 - I$

E| E → F → G → H → I| I $E=4+4=8 - I$

T| T → U → V → W → X| X $T=19+4=22 - X$

A| A → B → C → D → E| E $A=0+4=4 - E$

T| T → U → V → W → X| X $T=19+4=22 - X$

N| N → O → P → Q → R| R $N=13+4=17-R$

O| O → P → Q → R → S| S $O=14+4=18 - S$

O| O → P → Q → R → S| S $O=14+4=18 - S$

N| N → O → P → Q → R| R $N=13+4=17-R$

(Spaces remain unchanged.)

Encrypted Message (Ciphertext)

QIIX EX RSSR

Verification:

Plaintext : M E E T A T N O O N

Ciphertext: Q I I X E X R S S R

Thus, after applying a Caesar Cipher with a shift of 4, the message "MEET AT NOON" becomes:

QIIX EX RSSR

(b) Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

The authorized member knows that the sender used a Caesar Cipher with a shift of +4. To recover the original message, the receiver performs the reverse operation by shifting every encrypted letter 4 positions backward in the alphabet.

This process is called decryption.

Decryption Steps:

FORMULA: $D = (C - K) \bmod 26$

Cipher Letter | Shift -4 | Plain Letter

Q | $Q \rightarrow P \rightarrow O \rightarrow N \rightarrow M$ | M

I | $I \rightarrow H \rightarrow G \rightarrow F \rightarrow E$ | E

I | $I \rightarrow H \rightarrow G \rightarrow F \rightarrow E$ | E

X| X → W → V → U → T| T

E| E → D → C → B → A| A

X| X → W → V → U → T| T

R| R → Q → P → O → N| N

S| S → R → Q → P → O| O

S| S → R → Q → P → O| O

R| R → Q → P → O → N| N

Recovered Message

MEET AT NOON

Working Principle

- The sender encrypts the message using a shift of +4.
- The receiver decrypts the message using a shift of -4.
- Since both sender and receiver know the same shift value, the original message can be recovered accurately.
- If an unauthorized person does not know the shift value, the message appears meaningless.

Advantages of Caesar Cipher:

1. Very easy to implement.
2. Fast encryption and decryption.
3. Useful for learning the basic concepts of cryptography.
4. Requires only a single secret key (shift value).

Limitations:

1. Only 25 possible shift keys, making it easy to break using brute-force attacks.
2. Vulnerable to frequency analysis.

3. Not suitable for protecting sensitive information in modern communication.

Conclusion

The Caesar Cipher successfully converts "MEET AT NOON" into "QIIX EX RSSR" using a shift of 4. An authorized receiver simply shifts each letter 4 positions backward to retrieve the original message. Although this method is simple and effective for educational purposes, it does not provide strong security for modern communication systems because it can be broken easily.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics

A) Scenario

A cybersecurity intern intercepts the coded message:

"GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

2(a) Deduce the most likely original message.

The intercepted message appears to be encrypted using the Atbash Cipher, which is a classical substitution cipher. In this cipher, each letter of the alphabet is replaced by its opposite letter.

Atbash Cipher Mapping

Plain Letter| Cipher Letter

A| Z

B| Y

C| X

D| W

E| V

F| U

G| T

H| S

I| R

J| Q

K| P

L| O

M| N

N| M

O| L

P| K

Q| J

R| I

S| H

T| G

U| F

V| E

W| D

X| C

Y| B

Z| A

Step-by-Step Decryption

Cipher Letter| Plain Letter

G| T

S| H

R| I

H| S

R| I

H| S

Z| A

H| S

V| E

X| C

I| R

V| E

G| T

Decrypted Message

THIS IS A SECRET

Hence, the most probable original message is:

"THIS IS A SECRET"

(b) Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics

The intercepted message is likely encrypted using the Atbash Cipher because of its unique substitution pattern.

Pattern Analysis

1. Symmetric Substitution

- Atbash replaces each alphabet letter with its opposite.
- Examples:

- A ↔ Z
- B ↔ Y
- C ↔ X
- D ↔ W
- E ↔ V

2. Repeating Word Pattern

- The encrypted word "RH" appears twice.
- After decryption, it becomes "IS", which is a common English word.

3. Single-Letter Word

- The single-letter word "Z" decrypts to "A".
- In English, A is the most common one-letter word.

4. Meaningful Sentence

- After decryption, the message becomes "THIS IS A SECRET", which is grammatically correct and meaningful.
- This confirms that the cipher has been correctly identified.

5. Simple Classical Cipher

- Atbash is a monoalphabetic substitution cipher.
- It uses a fixed substitution table instead of a secret key.
- Since there is no changing key, anyone familiar with the Atbash mapping can decrypt the message.

Advantages of Atbash Cipher

- Very simple to understand and implement.
- No key is required for encryption or decryption.
- Useful for learning the concept of substitution ciphers.

Limitations

- Provides very weak security.
- The substitution pattern is fixed and publicly known.
- Easily broken using frequency analysis or by recognizing common word patterns.
- Not suitable for protecting confidential information in modern communication systems.

Conclusion

The ciphertext "GSRH RH Z HVXIVG" is decrypted using the Atbash Cipher. By replacing every letter with its opposite alphabet letter, the original message is obtained as:

"THIS IS A SECRET"

The repeated word pattern, single-letter word, and meaningful English sentence strongly justify that Atbash is the correct decryption method. Although it is useful for educational purposes, it is not secure enough for modern cyberse

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- Demonstrate how the encryption process transforms the message step-by-step.
- Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- Analyze how investigators could detect the presence of hidden data within the image.
- Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

